CHAPTER 01

Enterprise Architecture
Introduction, Purpose and Different Elements

holistic, umbrella view

what is enterprise architecture

focuses on capturing the bigger picture together

- **Enterprise architecture** (**EA**) is "a well-defined practice for conducting enterprise analysis, design, planning, and implementation, using a holistic approach at all times, for the successful development and execution of strategy".

- Enterprise Architecture applies architecture principles and practices to guide organizations through the business, information, process, and technology changes necessary to execute their strategies. These practices utilize the various aspects of an enterprise to identify, motivate, and achieve these changes."

★ understand what is classified under each type of principle

EA principles

✓ multi-faceted and focuses on different branches of principle

- **General principles**

1. IT and business alignment
2. Maximum benefits at the lowest cost and risk
3. Business continuity
4. Compliance with standards and policies
5. Adoption of the best practices for the market

- **Information principles**

6. Information treated as an asset
7. Shared information
8. Accessible information
9. Common terminology and data definitions
10. Information security

EA principles

- **Application principles**

1. Technological independence
2. Easy-to-use applications
3. Component reusability and simplicity
4. Adaptability and flexibility
5. Convergence with the enterprise architecture
6. Enterprise architecture also applies to external applications
7. Low-coupling interfaces → loose coupling
8. Adherence to functional domains

- **Technology principles**

9. Changes based on requirements
10. Control of technical diversity and suppliers
11. Interoperability

what is enterprise architecture(cont)

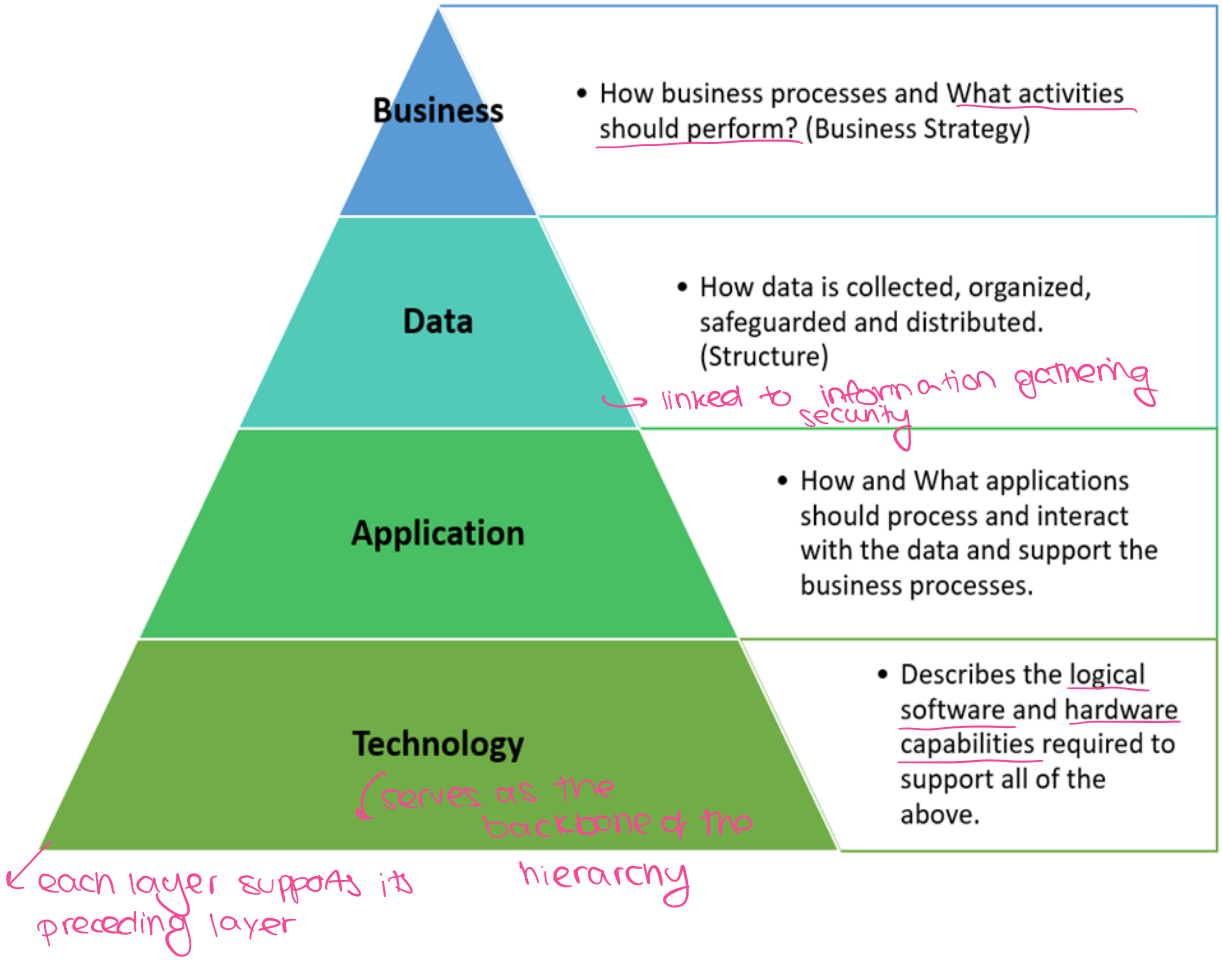
- With regard to resources, one of the greatest challenges that many enterprises continue to face is how to identify the business and technology components of strategic initiatives. difficult to fuse the together
- A big part of this challenge is that technology, information technology (IT) in particular, has historically not been viewed as a strategic asset. As such, planning activities often have focused on the development of individual technology solutions to meet particular organizational requirements.
- The following equation is the 'sound bite' version of what EA is all about and helps to differentiate between EA and other types of IT planning… that EA is driven by strategic goals and business requirements.

EA = S + B + T

- *Enterprise Architecture = Strategy + Business + Technology*

Components of Enterprise Architecture

"Defines the business, the information necessary to operate that business, the technologies necessary to support the business operations, and the transitional processes necessary for implementing new technologies in response to the changing needs of the business."



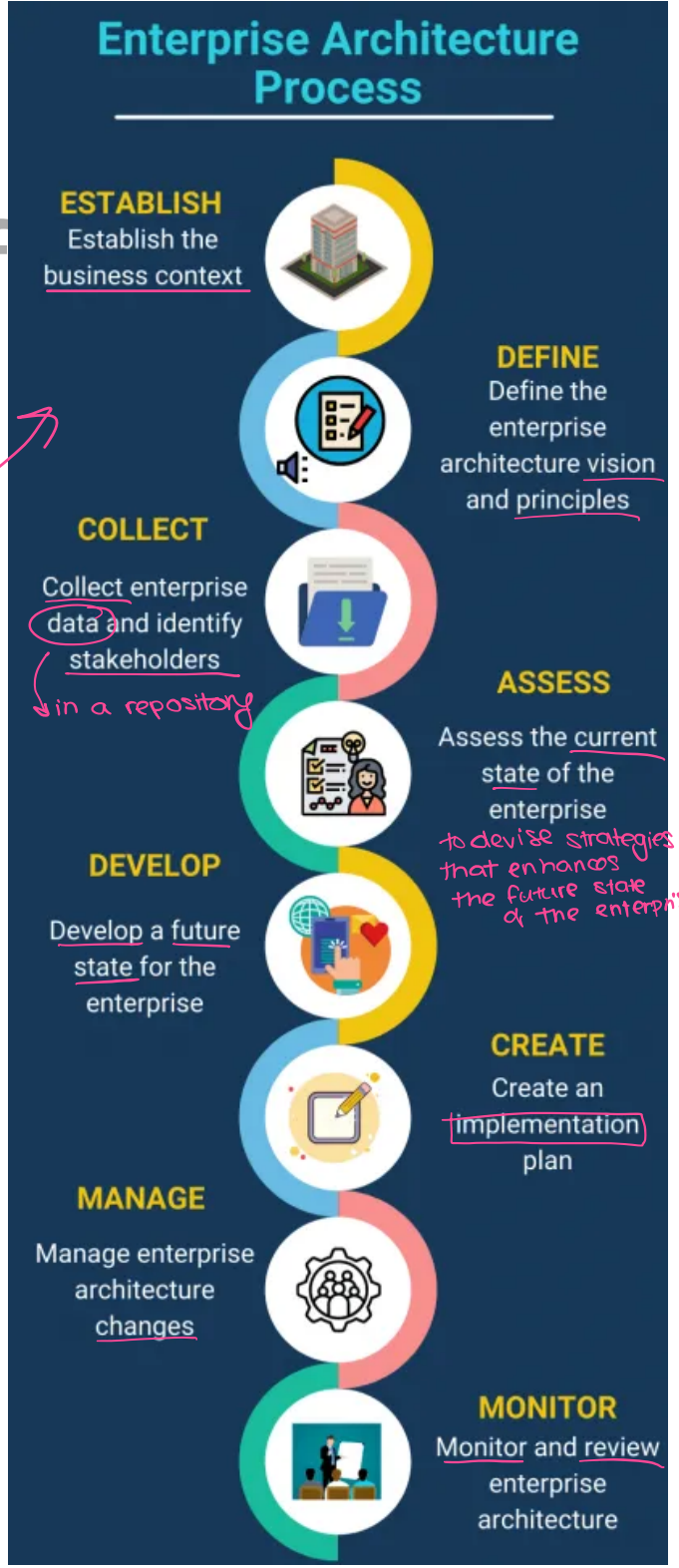
Benefits of EA

companies implementing EA concepts successfully:
1. SAMA
2. aramco
3. riyadh air
4. stc

- A blueprint for the enterprise that can be used to align IT with business goals
- A way to optimize enterprise performance
- A way to manage enterprise information systems
- A way to improve enterprise communication and collaboration
- A way to facilitate enterprise change management processes
- A way to enhance enterprise innovation is by understanding the enterprise as a whole system.

What is the process of enterprise architecture?

agile, continuous cycle that follows a process as shown



Enterprise Architecture as a Meta - Discipline

- An enterprise-wide architecture should serve as an authoritative reference, source of standards for processes / resources, and provider of designs for future operating states. An EA is therefore the architecture of the enterprise and should cover all elements and aspects.
- It also resolves the "battle of best practices" and competition between sub-architectural domains which can be problematic for organizations that are trying to become for efficient.

Enterprise Architecture as a Meta – Discipline(cont)

- EA is holistic and serves as an umbrella or "meta-context" for all other management and technology best practices. The EA also creates abstract views, analyses, and models of a current or future enterprise that helps people make better plans and decisions.
- EA extends beyond technology planning, by adding strategic planning as the primary driver of the enterprise, and business planning as the source of most program and resource requirements.

T/F-focus of EA is limited to technology -False

- When EA is the architecture of an organization in all dimensions, it becomes the over-arching, highest level discipline and the authoritative reference for standards and practices

companies with strong infrastructure are willing to

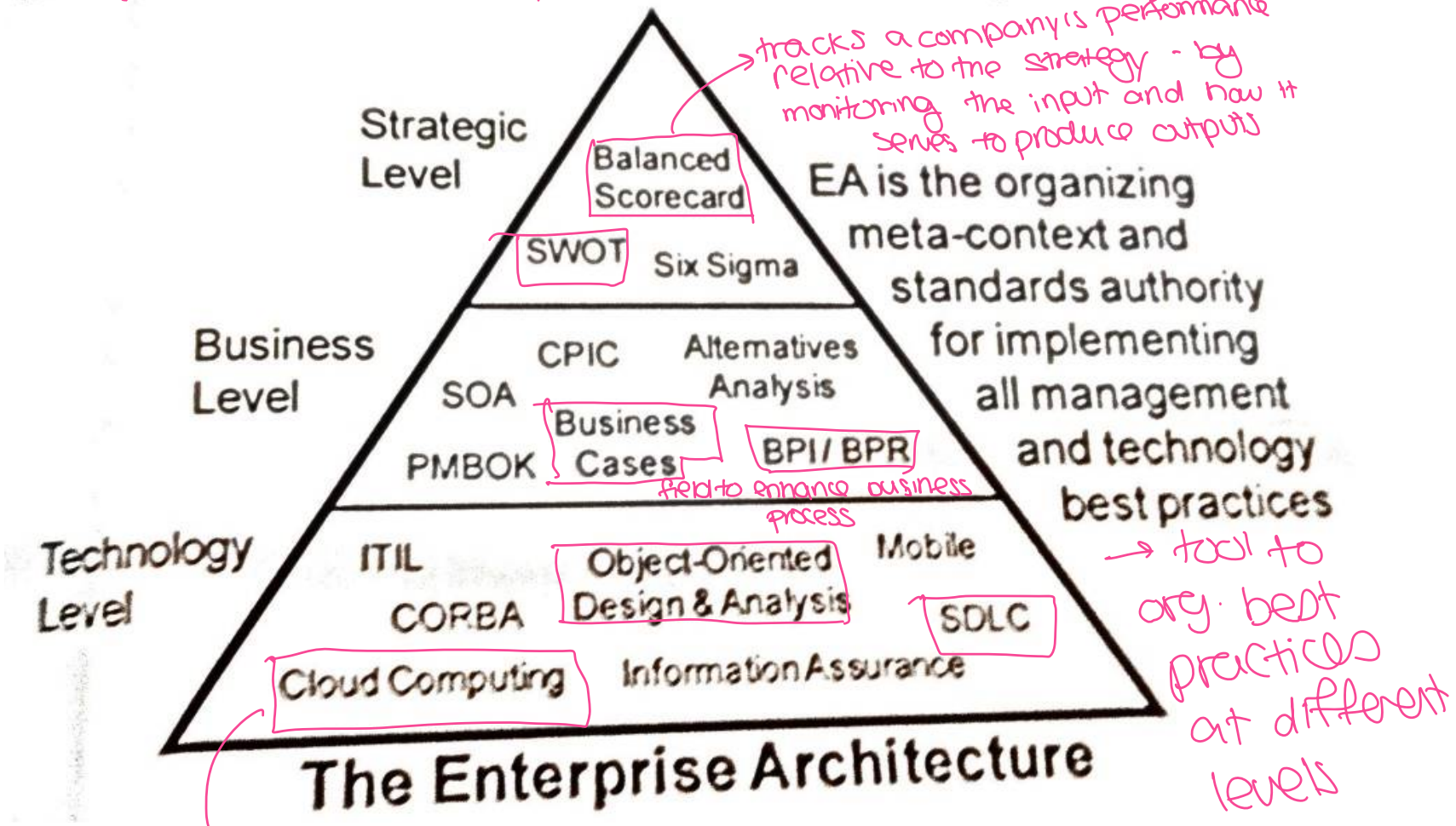
introduce EA framework in order to improve their

EA as the over-arching and highest level discipline

Strategy

A review footnotes for details

Below Figure illustrates how EA serves as an organizing context for the adoption and use of best practices.



The Enterprise Architecture Approach

For an EA approach to be considered to be complete, the six core elements€
shown in Figure 1-4 must be present and work interactively together.

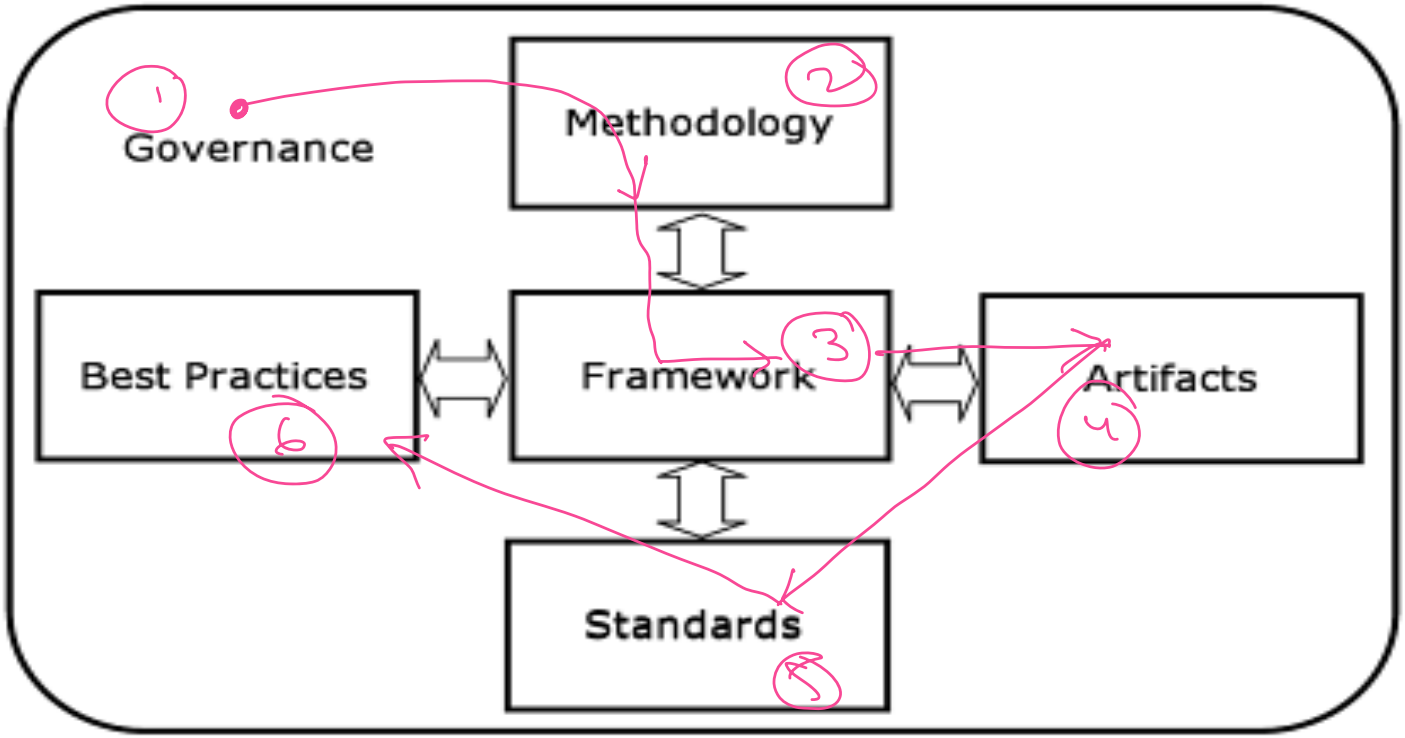


Figure 1-4: Core Elements of an Enterprise Architecture Approach

The Enterprise Architecture Approach (cont)

- **Governance**

  The first core element is "Governance" which identifies the planning decision-making, and oversight processes and groups that will determine how the EA is developed and maintained, accomplished as part of an organization's overall governance.

- **Methodology**

  The second core element is "Methodology" which are specific steps to establish and maintain an EA program, via the selected approach.

The Enterprise Architecture Approach (cont)

connects all domains within a business architecture

- 3 **Framework**

The third core element, which identifies the scope of the overall architecture and the type and relationship of the various sub-architecture levels and threads. Not all frameworks allow for sub-domains or are able to integrate strategy, business, and technology planning.

Three threads are security, standards and workforce. (skills)

↳ refers to the three S's: security, standards, skills (workforce)

Sub-architecture levels are strategic direction, business practices, information flows, systems and applications and technology resources.

- 4 **Artifacts**

The fourth core element, which identifies the types and methods of documentation to be used in each sub-architecture area including strategic analyses, business plans, internal controls, security controls. and models of workflow, databases, systems, and networks. This core element also includes the online repository where artifacts are stored.

↓documentation covering all sub-architecture domains

The Enterprise Architecture Approach (cont)

- **Standards**

  The fifth core element is "standards" which identify business and technology standards for the enterprise in each domain, segment, and components of the EA. This includes recognized international, national, local and industry standards as well as enterprise-specific standards.

- **Best Practices**

  The sixth core element is "Associated Best Practices" which are proven ways to implement parts of the overall architecture or sub-architectures, in context of the over-arching EA.

Enterprise Architecture Activities

- Enterprise architecture is accomplished through a **management program and an analysis and design method** that is repeatable at various levels of scope.

- Together the EA program and method provide an ongoing capability and actionable, coordinated views of an enterprise's strategic direction. business services, information flows, and resource utilization.

Enterprise Architecture Activities (cont)

EA as **a management Program:** An EA program is part of an overall governance process that determines resource alignment, develops standardized policy, enhances decision support, and guides development activities.

EA can help to identify gaps in the performance of line of business activities/programs and the capabilities of supporting IT services, systems, and networks.

- **Strategic Alignment**: Connects goals, activities, and resources
- **Standardized Policy**: Resource governance and implementation
- **Decision Support**: Financial control and configuration management
- **Resource oversight**: Lifecycle approach to development/management

Enterprise Architecture Activities (cont)

As **an analysis and design method**, EA provides:

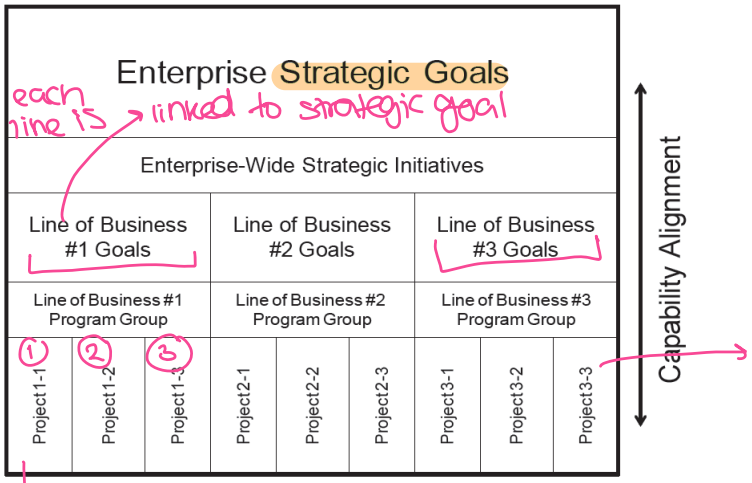
- **EA Approach**: The framework, analysis/design method, and artifact set
- **Current Views**: Views of as-is strategies, processes, and resources
- **Future Views**: Views of to-be strategies, processes' and resources
- **EA Management Plan**: A plan to move from the current to the future EA

EA As a management Program (cont)

1. Strategic Alignment

EA supports strategic planning and other operational resource planning processes by providing macro and micro views of how resources are to be leveraged in accomplishing the goals of the enterprise.

This helps to maximize the efficiency and effectiveness of these resources, which in turn will help to promote the enterprise's competitive capabilities.



Development projects within the enterprise should be reviewed to determine if they support one or more of the enterprise's strategic goals. If a resource and/or project is not aligned, then its value to

EA As **a management Program (cont)**

2. Standardized Policy

EA supports the implementation of standardized management policy pertinent to the development and utilization of IT and other resources.

By providing a holistic, hierarchical view of current and future resources, EA supports the establishment of policy for:

- *Identifying strategic and operational requirements ,*
- *Determining the strategic alignment of activities and resources*
- *Developing enterprise-wide business and technology resources*
- *Prioritizing the funding of programs and projects*
- *Overseeing the management of programs and projects*
- *Identifying performance metrics for programs and projects*
- *Identifying and enforcing standards and configuration management*

EA As a management Program (cont)

3. Decision Support

- EA provides support for IT resource decision-making at the executive, management, and staff levels of the enterprise.

1. **At the executive level**, EA provides visibility for large IT initiatives and supports the determination of strategic alignment.
2. **At the management level**, EA supports design and configuration management decisions, as well as the alignment of IT initiatives with technical standards for voice, data, video, and security.
3. **At the staff level**, EA supports decisions regarding operations, maintenance, and the development of IT resources and services.

Example: at unit decided to make a new system

EAs a management Program (cont)

4. Resource Oversight

- EA supports **standardized approaches** for overseeing the development of capabilities and optimizing supporting resources. Depending on the scope of the resources involved and the available timeframe for development, various system development lifecycle methods can be used to reduce the risk that cost, schedule, or performance parameters may not be met.

- EA further supports standardized, **proven approaches** to project management that promote the comprehensive and effective oversight of ongoing programs and new development projects.

- Finally, EA supports the use of **a standardized process** for selecting and evaluating investment in IT resources from a business and financial perspective.

EA aims to standardize process / policy across diff dep and domains within the enterprise

As an analysis and design method

- As **an analysis and design method**
- **Note: Full detail is in Ch.4**
- http://eapad.dk/ea3-cube/overview/
- http://www.youtube.com/watch?v=rhrjueWLPmE
- http://www.youtube.com/watch?v=qDI2oF1bASk

Archi: Reading Material

- https://sparxsystems.com/resources/tutorials/archimate/index.html